

Analytica - Sankalp 11.0
ORGANIC-CHOICE ANALYTICS CHALLENGE

Decoding Indian Consumer Behaviour for
Premium Food Products — GreenBazaar



COMMUNITY UPDATES

Problem Statement

GreenBazaar, one of India's premier premium retail chains (50+ stores across **Tier-1, Tier-2, and Tier-3 cities, 500+ crore annual ₹ revenue**), is underperforming in the organic food segment. Organic food currently contributes only **4% of total sales, well below the industry benchmark of 12%, despite the Indian organic food market growing at a 25% CAGR and projected to reach 75,000 crore by ₹ 2027**. This underperformance — driven by misallocated shelf space, generic marketing, poor inventory decisions, and unjustified premium pricing — **represents an estimated 50 crore in lost annual revenue**.

Business Question: Which customers are drawn to premium organic food, and how can GreenBazaar use that understanding to better target marketing, curate shelf space, and manage inventory?

Brief Summary

PHASE DESCRIPTION

This analysis was conducted through five key phases:

- **Phase 0** — Data Cleaning & Preparation: Checked for missing values and duplicates, and treated outliers using the IQR method.
- **Phase 1** — Exploratory Data Analysis: Analyzed and visualized customer distributions, spending patterns, and purchase intent across demographic, economic, and lifestyle variables.
- **Phase 2** — Feature Engineering & Selection: Created business-relevant features and identified the strongest drivers of customer spending and purchase intent.
- **Phase 3** — Business Problem-Solving Framework: Developed K-Means customer segments and mapped each segment to specific shelf, marketing, pricing, and inventory strategies.
- **Phase 4** — Insights & Recommendations: Converted the analysis into data-driven insights and prioritized recommendations to address the ₹50 crore organic revenue gap.
- **Phase 5** — Bonus: Predictive Modeling: Built a stacking ensemble using Random Forest, XGBoost, CatBoost, LightGBM, and Gradient Boosting, with Ridge as the meta-model, to predict Purchase_Intent

PHASE 0 : STEP 1

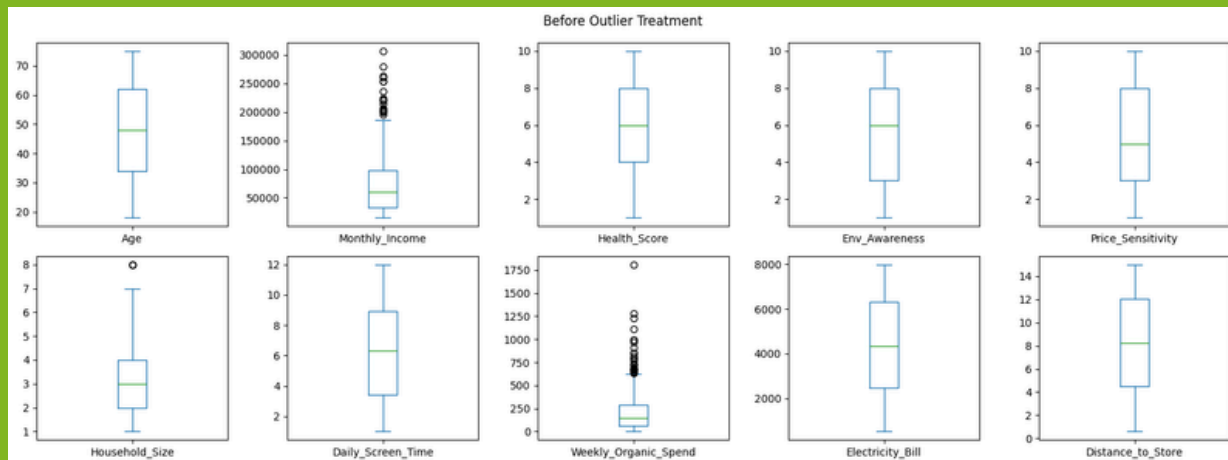
Checking Missing & Duplicate Data

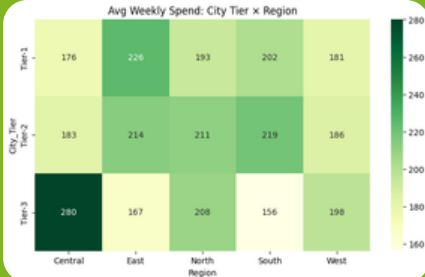
The dataset was checked for both missing values and duplicate records during the data cleaning process. The analysis confirmed that **no missing values and no duplicate records** were found in the dataset.

PHASE 0 : STEP 2

Outlier Detection & Treatment

Outliers were detected in key numerical variables using the **IQR (Interquartile Range) method**. Values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ were treated as extreme values. Instead of removing these records, the **outliers were capped at the lower and upper IQR bounds**, preserving valuable customer information while reducing the impact of extreme values on further analysis.





Avg. Weekly Organic Spend : City Tier × Region

- Cross-tabulates spend by both city tier and region at once, giving **15 specific combinations** instead of two separate charts.
- Reveals that **Tier-2 Central (₹280) and Tier-1 East (₹226)** are the true spending hotspots — not simply "Tier-1" or "any metro" as a whole.
- More precise than looking at City Tier or Region alone, since it shows spend can vary sharply even within the same tier depending on region.

Avg Weekly Organic Spend by All Categorical Features

- One combined figure with a bar chart per categorical column (Gender, Education, Marital Status, City Tier, Region, etc.), each ranked by average spend.
- Lets you scan every **demographic and lifestyle category** at once to spot which groups spend the most, without opening 11 separate charts.
- Serves as the categorical counterpart to the numeric correlation heatmap — showing **where spend differences exist that a numeric-only analysis would miss**.



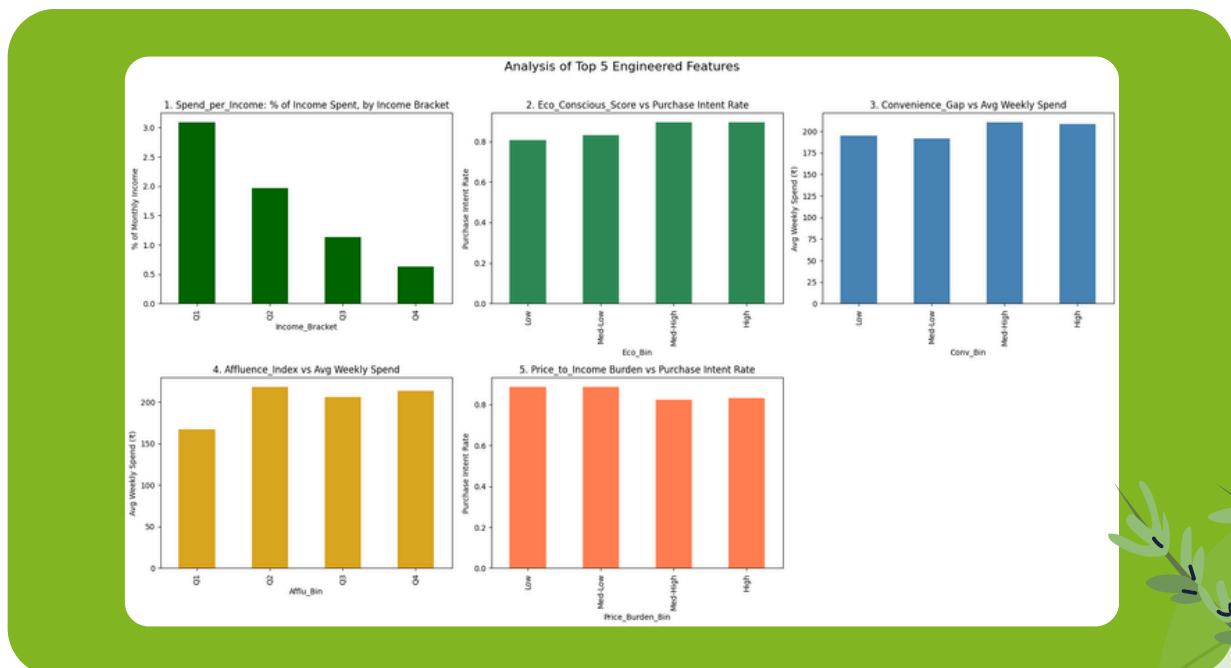
Purchase Intent Rate by All Categorical Features

- Same layout as the spend chart above, but shows the % of **customers who say "Yes" to Purchase Intent** within each category instead of ₹ spent.
- Lets you separate **"who wants organic" from "who actually pays for it"** — a customer group can rank high on intent but low on spend, or vice versa.
- Critical for validating Task 4's core insight: intent is high almost everywhere (82-88%), so this chart is the visual proof that desire isn't GreenBazaar's real bottleneck.

Engineered Features	Business Meaning
Spend_per_Income	Weekly organic spend as a % of monthly income – reveals budget priority, not just affordability
Eco_Conscious_Score	Health + Environmental Awareness minus Price Sensitivity – captures values driven buying
Convenience_Gap	Distance to store × non-online-shopper flag – captures real access friction
Affluence_Index	Composite wealth score from income, cars, credit cards, and electricity bill
Price_to_Income	Price sensitivity relative to per-person income – separates real budget constraint from attitude

Feature selection combined correlation ranking, mutual information, and Random Forest importance. **Spend_per_Income emerged as the single strongest engineered driver** of both purchase intent and segment membership, followed by Eco_Conscious_Score and Convenience_Gap — these three features formed the backbone of the Task 3 segmentation model.

Engineered Features Relationships

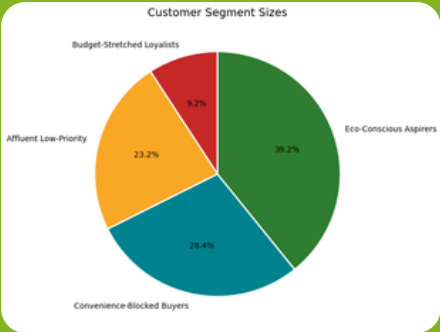


Segment Profile

Segment	Size	Avg. Income (₹)	Avg. Spend (₹)	Intent Rate
Eco-Conscious Aspirers	196	54,090	172	88%
Convenience-Blocked Buyers	142	56,735	214	82%
Affluent Low-Priority	116	136,028	197	86%
Budget-Stretched Loyalists	46	19,755	301	87%

Segment-to-Action Framework

Segment	Shelf Placement	Pricing Strategy	Inventory Decision
Budget-Stretched Loyalists	Guarantee availability of value-pack items	Smaller packs + loyalty discounts	Prioritize affordable core SKUs
Affluent Low-Priority	Prominent premium placement near checkout	Standard-to-premium pricing with bundling	Stock wide, exclusive premium SKU range
Convenience-Blocked Buyers	Prominent premium placement near checkout	Standard pricing + discounted delivery	Prioritize delivery/online-channel stock
Eco-Conscious Aspirers	Prime, easy-access placement near entrance	Introduce accessible mid-tier organic line	Increase entry-level SKU variety



Customer Segment Sizes

Key Insights

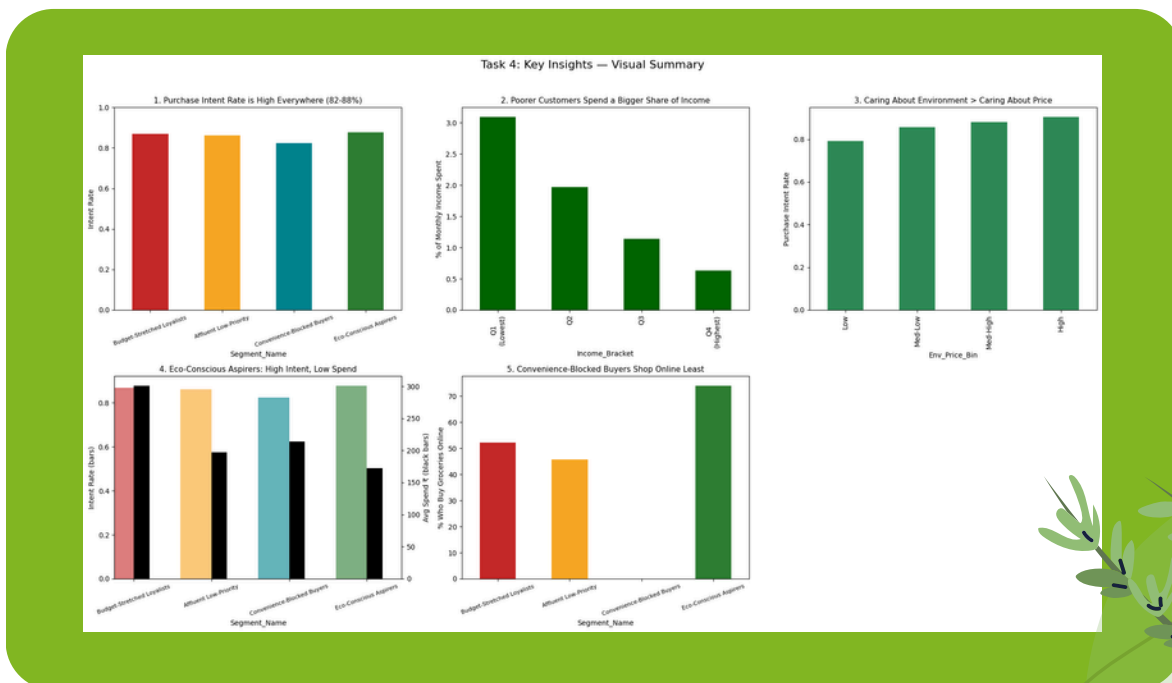
- **Purchase intent is high (82–88%) across all segments**, indicating that conversion, not demand is the main challenge.
- **Lower-income customers spend more** of their income on organic, **at 3.4% versus 0.6% for the highest-income group**.
- **Health and environmental consciousness drives intent more** strongly than price sensitivity (**90% vs. 79% conversion**).
- **Eco-Conscious Aspirers are the largest segment** (196) with 88% intent but only ₹172 weekly spend, **creating the biggest revenue opportunity**.
- Convenience-Blocked Buyers (142) face access **barriers due to store distance** and **limited online grocery adoption**.

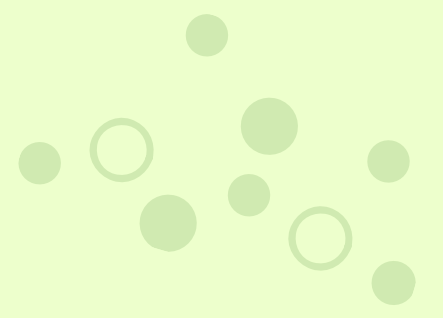
Major Factors GreenBazaar Should Focus On

- **Conversion over awareness**: Most customers already show purchase intent, so the **focus should be on converting intent into actual purchases**.
- **Value communication over price cuts**: Health and environmental consciousness drives stronger conversion, making better value communication more effective than **blanket discounts**.
- **Geographic reallocation**: **Tier-2 customers spend more on organic products than Tier-1 customers**, suggesting a need to reconsider metro-focused investments.
- **Access and convenience**: **Store distance and limited online shopping options** are important barriers for convenience-blocked customers.

Which Customers to Target — Priority Order

- **Eco-Conscious Aspirers (196)**: Largest segment with the highest intent but lowest spend — **the biggest revenue opportunity**; introduce a mid-tier product line.
- **Convenience-Blocked Buyers (142)**: Purchase intent is limited by access barriers — **invest in delivery and online shopping options**.
- **Affluent Low-Priority (116)**: High income but lower organic priority — **use premium positioning** and bundling to increase spending.
- **Budget-Stretched Loyalists (46)**: Smallest segment but highly committed — protect their access through **value packs and loyalty pricing**.





(Pranav Mishra & Arhan Ansari)

TEAM : THINK CORE

Thank you For Your Time and Attention

NOTE: The Jupyter Notebook (.ipynb) contains additional graphs and phases that could not be included in the PDF due to the 5-page limit. Kindly refer to the notebook as well for the complete analysis, visualizations, and results.

